

Cross-Geo Product Matching and Grouping Using Multimodal Embeddings and Graph Clustering

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Abstract

We address the problem of identifying identical fashion products across 13 European markets, where the same item appears with different titles, prices, and images. Our system combines multilingual sentence embeddings [5], CLIP image embeddings [6], and structured meta-data through a two-phase pipeline. Phase 1 classifies predefined groups using LightGBM [7] on aggregated pairwise features ($F1 = 0.9745$). Phase 2 performs open-world product clustering via FAISS [9] approximate nearest neighbors, a trained pair-level edge scorer, contrastive fine-tuning, and Leiden [4] graph clustering, achieving pairwise $F1 = 0.9005$ on items_phase1. Inspired by the fashionID architecture [2], we train a multimodal projection that fuses text, image, and price signals into a shared 128-dimensional metric space. An ablation study confirms that each component contributes independently, with the full pipeline improving pairwise $F1$ by +0.09 over the base embedding baseline.

1. Introduction

Product matching is a key capability for e-commerce platforms, enabling de-duplication, competitive intelligence, and recommendation of substitutes [2]. GLAMI, a fashion search engine operating across 13 European markets with over 20 million items, faces this challenge at scale: the same physical product appears under different titles (in different languages and scripts), different prices (in local currencies from EUR to HUF to BGN), and often different images. As Toth et al. [2] observe, fashion product matching is more similar to person re-identification than matching other product categories, because visual information is highly important and products are rarely identified through unique names beyond brand names.

The NI-ADM 2026 competition formalizes this as two tasks. Phase 1 is **binary classification**: given 15,000 groups of 5 items, predict whether each contains at least one duplicate pair (evaluated by $F1$). Phase 2 is **open-world entity resolution**: given $\sim 200k$ unlabeled items, construct disjoint product groups (evaluated by pairwise $F1$). Three challenges make this problem difficult: (a) **multimodality** - text, images, price, and categorical metadata must be fused effectively; (b) **cross-geo variation** - $\sim 40\%$ of product labels span 2+

markets with titles in completely different languages; (c) **class imbalance** — in Phase 1, the 80/20 negative/positive group ratio requires careful threshold calibration; in Phase 2, the pair-level imbalance is extreme ($\sim 0.002\%$ of all pairs are true matches), making brute-force scoring of ~ 20 billion pairs both computationally infeasible and statistically degenerate. Candidate blocking (FAISS) and class-weighted training are essential.

2. Related Work

Entity matching. Christophides et al. [10] survey end-to-end entity resolution for big data, identifying blocking, matching, and clustering as the three core stages. Peeters et al. [1] demonstrate that LLMs achieve very high precision but at costs prohibitive for millions of items. Li et al. [11] show that pre-trained language models fine-tuned as binary classifiers on concatenated text pairs achieve state-of-the-art on textual entity matching benchmarks. Peeters and Bizer [12] extend this to contrastive learning, demonstrating more efficient use of training data than cross-encoder approaches.

Multimodal embeddings. Our text encoder builds on Sentence-BERT [5], which adapts pre-trained transformers to produce fixed-size sentence embeddings via siamese networks. The multilingual variant supports 50+ languages, critical for cross-geo matching. For images, we use CLIP [6], which learns a shared text-image embedding space through contrastive pre-training on 400M pairs. Toth et al. (Zalando paper) [2] find that CLIP encoders surprisingly outperform DINOv2 [13] on visual product matching despite DINO’s image-only self-supervised training, and that image information dominates text for fashion matching. For scalable retrieval, Johnson et al. [9] introduce FAISS with IVF indexing for billion-scale approximate nearest neighbor search.

The fashionID architecture. Most relevant to our work is Toth et al. [2], who propose fashionID: a late-fusion encoder that concatenates pooled CLIP image embeddings, CLIP text embeddings, and a 3-dimensional numeric vector (price, log-price, number of sizes), then projects through a single linear layer into a 192-dimensional metric space. Training uses supervised contrastive loss [14] with large batch sizes (16k) and batch sampling that maximizes positive pairs. A

key design decision is freezing the pre-trained encoders and training only the linear projection (0.6M parameters), enabling training in under 10 minutes on a single GPU. Their ablation shows that numeric features contribute ~ 1 AUCPR point. However, their approach has **limitations** in our setting: (a) they rely solely on embedding distance without leveraging structured metadata like department or color IDs; (b) they use brand-based blocking for retrieval, which is unavailable in the GLAMI dataset; (c) a single cosine threshold cannot distinguish same-category products that have high semantic similarity but are distinct items. We address these through a two-stage pipeline where FAISS provides candidates and a LightGBM classifier incorporates structured features for final edge decisions.

Graph clustering. Traag et al. [4] introduce the Leiden algorithm, which improves on Louvain by guaranteeing internally connected communities. This property is important for product grouping, where a single weak transitive edge in Louvain can mistakenly merge distinct product clusters.

3. Methodology

3.1. Data Preparation

The dataset comprises 928k labeled training items and ~ 200 k unlabeled items per phase across 13 markets. We parse departmentIds to integer sets and colorTagIdsString to integer sets for Jaccard computation. Prices are normalized to EUR using fixed exchange rates (CZK: 0.04, HUF: 0.0025, BGN: 0.5112, RON: 0.2, PLN: 0.2326). EDA confirmed 91.9% of product labels share identical departmentIds across items; brandEditionTagId was dropped (99.7% missing). Following [2], we build a numeric feature vector from log-normalized EUR price.

3.2. Feature Extraction

Text embeddings. Each item’s title and truncated description (500 chars) are encoded using *paraphrase-multilingual-mpnet-base-v2* [5] (768-dim, 50+ languages), L2-normalized.

Image embeddings. Product images are encoded using CLIP ViT-B/32 [6] (512-dim), with coverage varying by market; items without images are masked out in the projection via a has_image flag. Following [2], who find that CLIP provides a stronger visual baseline than DINOv2, we use CLIP as our image encoder throughout.

Contrastive fine-tuning. We fine-tuned the text encoder with MultipleNegativesRankingLoss [12] on ~ 450 k positive pairs (same-label items) for 3 epochs (batch size 64, warmup ratio 0.1), shifting embeddings from generic semantic similarity toward product identity. For each anchor-positive pair (a_i, p_i) , all other positives in the batch serve as in-batch negatives:

$$\mathcal{L}_{\text{MNR}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\cos(a_i, p_i))}{\sum_{j=1}^N \exp(\cos(a_i, p_j))} \quad (1)$$

3.3. Phase 1: Group Classification

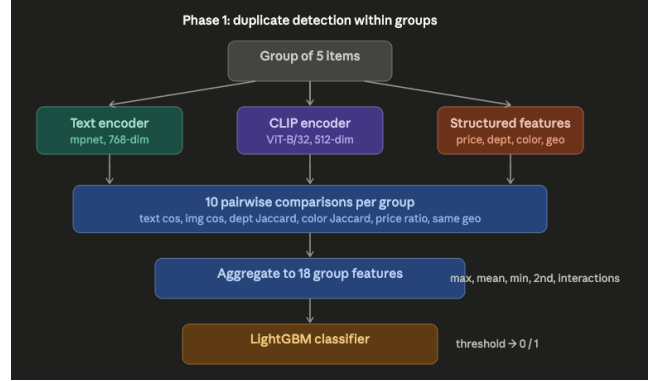


Figure 1: Phase 1 pipeline: pairwise feature extraction and LightGBM group classification.

For each group of 5 items, we compute all 10 pairwise comparisons, extracting per pair: text cosine similarity, image cosine similarity, department Jaccard, color tag Jaccard, EUR price ratio (min/max), and same-geo flag. These are aggregated into several group-level features (max, mean, min, 2nd-highest for embedding similarities; max, mean for structured features; plus interaction terms). A LightGBM [7] classifier (300 trees, depth 5, lr 0.05, scale_pos_weight=4.0 for the 80/20 class imbalance) is trained with 5-fold stratified cross-validation on 15,000 synthetic validation groups.

3.4. Phase 2: Product Clustering

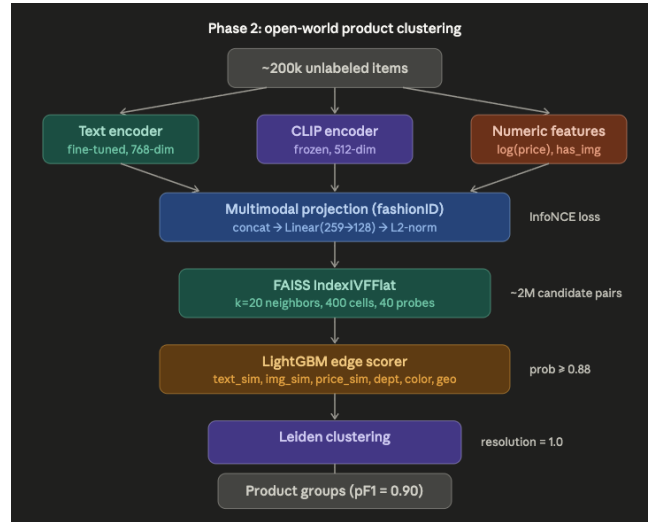


Figure 2: Phase 2 pipeline: FAISS candidate generation, LightGBM edge scoring, and Leiden graph clustering.

Phase 2 requires clustering ~ 200 k items into product groups, evaluated by pairwise F1. Our pipeline follows the two-stage approach of [2] (encoding + retrieval) but extends it with learned edge scoring and graph-based clustering:

(1) Candidate generation. FAISS IndexIVFFlat [9] (400 Voronoi cells, 40 probes) finds top-20 near-

est neighbors per item by inner product, reducing the search space from ~ 20 billion to $\sim 2\text{M}$ pairs.

(2) Edge scoring. A pair-level LightGBM (500 trees, depth 6) scores each candidate using 6 features: text_sim, img_sim, price_sim, dept_jaccard, color_jaccard, same_geo. This replaces the raw cosine threshold used by [2], allowing incorporation of structured metadata signals.

(3) Leiden clustering. A weighted NetworkX graph is built from scored edges. Leiden [4] community detection identifies product groups with guaranteed internal connectivity.

(4) Post-processing. Clusters exceeding 150 items are recursively re-clustered at higher resolution.

(5) Confidence estimation. Each cluster is assigned a confidence score equal to the mean LightGBM match probability across all internal edges. Empirically, clusters with mean edge weight > 0.95 achieve $\sim 95\%$ purity, while lower-confidence bins (0.85–0.95) drop to 67–71%, reflecting transitive false positives introduced by Leiden.

3.5. Multimodal Projection

Inspired by fashionID [2], we train a projection head that fuses all modalities into a 128-dim metric space. The architecture: text_proj (768 $\rightarrow$ 128, ReLU), img_proj (512 $\rightarrow$ 128, ReLU), numeric (log-normalized price, has_image flag, zero-padded third dimension), fusion (259 $\rightarrow$ 128), L2-normalized. Unlike [2] who use a single linear layer with 192-dim output, we use ReLU activations and a smaller output dimension. Training uses InfoNCE loss with temperature $\tau = 0.07$ and batch sampling that maximizes positive pairs per batch (256 groups, 4 items per group). Let $\mathbf{z}_i$ denote the L2-normalized output embedding and y_i the product label; the loss averages over all anchors $\mathcal{A}$ that have at least one positive in the batch:

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} \log \frac{\sum_{j \neq i, y_j = y_i} \exp(\mathbf{z}_i^\top \mathbf{z}_j / \tau)}{\sum_{k \neq i} \exp(\mathbf{z}_i^\top \mathbf{z}_k / \tau)} \quad (2)$$

We also experimented with joint training of the text encoder and projection head simultaneously, where the encoder learns representations that complement the frozen CLIP image signal.

4. Baselines

We compare our method against baselines for both phases. **For Phase 1: B1 (Embedding threshold):** Thresholding max pairwise cosine similarity of multilingual text embeddings at 0.75 ($F1 = 0.900$). This handles cross-lingual matching via the multilingual encoder but ignores price, image, and metadata.

B2 (LightGBM, text + structured): LightGBM on 13 text and structured features without image embeddings ($F1 = 0.914$). Adding CLIP image features yields $F1 = 0.964$ (our full Phase 1 model).

For Phase 2: B3 (Base embeddings + LightGBM scorer): FAISS with a LightGBM edge scorer on paraphrase-multilingual-mpnet base embeddings, Leiden clustering (pairwise $F1 = 0.810$). The LightGBM scorer and graph clustering pipeline are fixed across all Phase 2 rows; subsequent improvements come exclusively from better embeddings.

5. Experiments

5.1. Validation Strategy

For Phase 1, we constructed 15,000 synthetic validation groups from labeled training items (3,000 positive, 12,000 negative), matching the 20/80 ratio. 5-fold stratified CV provides reliable estimates. For Phase 2, we sample 200k items from training data and evaluate pairwise $F1$ against ground-truth labels. The competition evaluates Phase 2 on a blind test set (items_phase_2.csv).

5.2. Hyperparameter Selection

Phase 1 LightGBM: 300 trees, depth 5, lr 0.05, 31 leaves, scale_pos_weight=4.0. Classification threshold swept in 0.01 steps to maximize $F1$. For Phase 2, grid search over k in $\{10, 20, 30\}$, edge threshold in $\{0.80, 0.85, 0.90\}$, and Leiden resolution in $\{0.8, 1.0, 1.2\}$ selected $k=20$, resolution=1.0. The edge scorer LightGBM threshold was tuned on held-out pairs (best at 0.88). Contrastive fine-tuning used 3 epochs, batch size 64, warmup ratio 0.1. The projection head was trained with temperature 0.07, cosine annealing LR schedule, and 5 epochs.

5.3. Quantitative Results

Table 1: Results across both phases. Phase 1 metrics from 5-fold CV (rows 1-3) and leaderboard (row 4). Phase 2 metrics from the competition server. P/R unavailable for server-evaluated submissions.

Phase	Method	F1	P	R
1	B1: Text emb. threshold (0.75)	0.900	0.910	0.889
1	B2: LightGBM (text + struct.)	0.914	0.945	0.886
1	Full: LightGBM (text+img+struct.)	0.964	0.989	0.941
1	Leaderboard (submitted)	0.9745	-	-
2	B3: Base emb. + LightGBM scorer	0.810	-	-
2	Contrastive FT text emb. + LightGBM scorer	0.883	-	-
2	Multimodal projection + LightGBM scorer	0.901	-	-
2	Top of leaderboard at the time of reporting	0.942	-	-

6. Results

6.1. Leaderboard Performance

Phase 1 leaderboard $F1 = 0.974$. Phase 2 pairwise $F1 = 0.9005$, with a gap of ~ 0.04 to the top score of 0.9424 in the time of reporting. The Phase 2 progression from baseline (0.810) to final submission (0.9005)

demonstrates the cumulative effect of contrastive fine-tuning (+0.073, with the LightGBM scorer and Leiden pipeline held fixed) and the multimodal projection (+0.018).

6.2. Ablation Study

Phase 1 ablation. Starting from text-only embedding threshold ($F1 = 0.900$): adding structured features (price, dept, color) with LightGBM raised $F1$ to 0.914 (+0.014); adding CLIP image embeddings raised $F1$ to 0.964 (+0.050). Feature importance: img_max (1041 splits), text x image interaction (745), emb_max (592), price_max (566). Notably, dept_max has near-zero importance in Phase 1 due to low variance (Jaccard mean = 0.997), but department remains valuable as a feature in the Phase 2 edge scorer.

Phase 2 ablation. The LightGBM edge scorer (which includes img_sim as a feature) and Leiden clustering pipeline are held fixed across all configurations; only the input embeddings change. The base text embeddings yield pairwise $F1 = 0.810$. Replacing them with contrastively fine-tuned embeddings (MNR loss, $\sim 450k$ positive pairs, 3 epochs) improves $F1$ to 0.8828 (+0.073), demonstrating that domain-specific fine-tuning toward product identity is the single largest gain. Replacing text-only embeddings with the multimodal projection (fusing text, image, and price into 128-dim via InfoNCE) yields $F1 = 0.9005$ (+0.018): in the previous step image information entered only as a secondary scorer feature; the projection promotes it into the primary retrieval space, allowing FAISS to surface better candidates.

6.3. Strengths and Weaknesses

Strengths. The pipeline successfully handles cross-lingual matching (40% of labels span 2+ geos). Currency normalization turns price into a strong discriminator. The modular architecture allows independent improvement of each stage. The two-stage scoring (FAISS + LightGBM) overcomes the limitation of [2]’s single-threshold approach by incorporating structured metadata.

Weaknesses. The gap to the leaderboard top (0.04 pairwise $F1$) suggests room for improvement. The two-stage training (encoder then projection) is suboptimal: the text encoder never sees the image signal during fine-tuning. Confidence calibration shows that high-confidence clusters (mean edge weight > 0.95 , 38k clusters) achieve $\sim 95\%$ purity, but lower-confidence bins drop to 67–71%, indicating that weak transitive edges remain a source of false positives. We use smaller pre-trained models (ViT-B/32) than [2]’s ViT-bigG-14, which limits visual feature quality.

7. Conclusion

We presented a multimodal product matching system that combines text, image, price, and categorical metadata for cross-geo fashion product identification. Key

findings: (1) image embeddings provide the largest single-component improvement in Phase 1 (+0.050 $F1$); (2) contrastive fine-tuning is the largest single improvement in Phase 2 (+0.07 pairwise $F1$); (3) the multimodal projection inspired by fashionID [2] yields a further +0.02 by fusing all signals; (4) a LightGBM edge scorer incorporating structured features (text, image, price, department, color, geo) enables lower FAISS candidate thresholds while maintaining precision, outperforming a raw cosine threshold; (5) Leiden clustering outperforms Louvain by producing tighter, internally connected communities.

Future work. (1) **Joint encoder + projection training:** currently the pipeline trains in two stages (fine-tune encoder, then freeze and train projection). Joint end-to-end training with InfoNCE would let the encoder learn representations that complement the image signal directly, potentially gaining 1-3 pairwise $F1$ points. (2) Scaling to larger pre-trained encoders (ViT-L or ViT-bigG as in [2]) could improve visual feature quality. (3) Brand name extraction from titles as an exact-match feature.

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